SYS 3060: Stochastic Decision Models
Homework 2 Solutions

Q1. Parking Garage with One Charging Station

Given:

$$\lambda = 4 \text{ vehicles/hour}, \quad \mu = 6 \text{ vehicles/hour}, \quad c = 5,$$

state space $\{0, 1, 2, 3, 4, 5\}$, and arrivals are blocked in state 5.

(a) CTMC representation (state-transition graph / generator)

This is a finite-capacity birth-death CTMC with birth rates

$$\lambda_n = \begin{cases} \lambda, & n = 0, 1, 2, 3, 4, \\ 0, & n = 5, \end{cases} \qquad \mu_n = \begin{cases} 0, & n = 0, \\ \mu, & n = 1, 2, 3, 4, 5. \end{cases}$$

A rate diagram (births to the right, deaths to the left):

$$\begin{tikzcd} 0 \arrow[r, leftrightarrow, "\lambda", "\mu"'] & 1 \arrow[r, leftrightarrow, "\lambda", "\mu"'] & 2 \arrow[r, leftrightarrow, "\lambda", "\mu"'] & 3 \arrow[r, leftrightarrow, "\lambda", "\mu"'] & 4 \arrow[r, leftrightarrow, "\lambda", "\mu"'] & 5 \end{tikzcd}$$

Equivalently, the generator Q has

$$q_{n,n+1} = \lambda \ (n = 0, \dots, 4), \quad q_{n,n-1} = \mu \ (n = 1, \dots, 5),$$

all other off-diagonals 0, and $q_{n,n} = -(\text{sum of rates out of } n)$.

(b) Leaving rates and next-step probabilities

(i) Rate at which the CTMC leaves state n:

$$\nu(n) = \begin{cases} \lambda, & n = 0, \\ \lambda + \mu, & n = 1, 2, 3, 4, \\ \mu, & n = 5. \end{cases}$$

(ii) Given that the CTMC leaves state n, the move probabilities are:
For $n = 1, 2, 3, 4$ (both moves possible),

$$\mathbb{P}(n \to n+1 \mid \text{leave } n) = \frac{\lambda}{\lambda + \mu}, \qquad \mathbb{P}(n \to n-1 \mid \text{leave } n) = \frac{\mu}{\lambda + \mu}.$$

For boundary states:

$$\mathbb{P}(0 \to 1 \mid \text{leave } 0) = 1, \qquad \mathbb{P}(5 \to 4 \mid \text{leave } 5) = 1.$$

(c) Steady-state probabilities π_n

Because this is a birth-death chain, detailed balance gives

$$\pi_n = \pi_0 \left(\frac{\lambda}{\mu}\right)^n, \qquad n = 0, 1, 2, 3, 4, 5.$$

Here $\lambda/\mu = 4/6 = 2/3$. Normalize:

$$1 = \sum_{n=0}^{5} \pi_n = \pi_0 \sum_{n=0}^{5} \left(\frac{2}{3}\right)^n = \pi_0 \cdot \frac{1-(2/3)^6}{1-2/3}.$$

Compute

$$\sum_{n=0}^{5} \left(\frac{2}{3}\right)^n = \frac{665}{243}, \qquad \Rightarrow \qquad \pi_0 = \frac{243}{665}.$$

Thus

$$(\pi_0, \pi_1, \pi_2, \pi_3, \pi_4, \pi_5) = \left(\frac{243}{665}, \frac{162}{665}, \frac{108}{665}, \frac{72}{665}, \frac{48}{665}, \frac{32}{665}\right).$$

(d) Blocking, busy station, and expected queue length

Blocking probability. By PASTA, the probability an arrival is blocked equals the steady-state probability the system is full:

$$\mathbb{P}(\text{blocked}) = \pi_5 = \frac{32}{665} \approx 0.0481.$$

Expected number of busy charging stations. There is only one charger. It is busy iff $N \geq 1$. Hence

$$\mathbb{E}[\text{busy chargers}] = \mathbb{P}(N \geq 1) = 1 - \pi_0 = 1 - \frac{243}{665} = \frac{422}{665} \approx 0.6346.$$

Expected number waiting in queue. With one server, the number waiting is $(N-1)^+$. Thus

$$\mathbb{E}[Q] = \sum_{n=0}^{5} (n-1)^+ \pi_n = \sum_{n=2}^{5} (n-1)\pi_n = \pi_2 + 2\pi_3 + 3\pi_4 + 4\pi_5.$$

Plugging values:

$$\mathbb{E}[Q] = \frac{108}{665} + \frac{144}{665} + \frac{144}{665} + \frac{128}{665} = \frac{524}{665} \approx 0.7880.$$

Q2. Two-Stage Charging with Blocking

Given:

$$\lambda = 6, \quad \mu_1 = 6, \quad \mu_2 = 6 \quad \text{(vehicles/hour)}.$$

Stage 1 capacity is 2 total (1 in service + 1 waiting). Stage 2 capacity is 2 total.

State: $X(t) = (I(t), J(t))$, with $J \in \{0, 1, 2\}$ the number at Stage 2 and

$$I \in \{0, 1, 2, 1b, 2b\}.$$

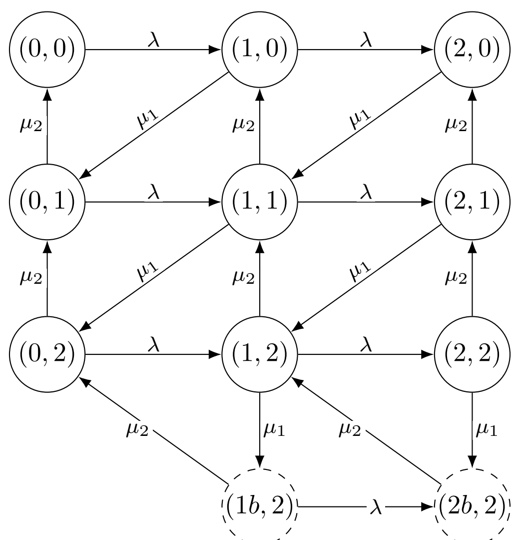
Blocked states only occur when $J = 2$.

Important modeling assumption

I assume that in state $1b$ (blocked EV at the Stage 1 charger and Stage 1 waiting spot empty), an external arrival *can* enter Stage 1 and occupy the waiting spot, sending the system to state $2b$. This matches the meaning of $2b$ as "blocked and also has one EV waiting behind."

All calculations below use that assumption.

(a) CTMC state-transition graph with rates



(b) Steady-state probabilities $\pi_{i,j}$

Solving the global balance equations for this finite CTMC yields the following stationary distribution.

Nonblocked states (i, j) with $i \in \{0, 1, 2\}$, $j \in \{0, 1, 2\}$:

	$j = 0$	$j = 1$	$j = 2$
$i = 0$	$\frac{7}{75}$	$\frac{7}{75}$	$\frac{2}{25}$
$i = 1$	$\frac{8}{75}$	$\frac{3}{25}$	$\frac{2}{25}$
$i = 2$	$\frac{14}{75}$	$\frac{2}{25}$	$\frac{1}{25}$

Blocked states:

$$\pi_{1b,2} = \frac{1}{25}, \qquad \pi_{2b,2} = \frac{2}{25}.$$

(These probabilities sum to 1.)

(c) Performance measures using π

Probability that external arrivals cannot enter Stage 1. External arrivals are blocked when Stage 1 is full: $I = 2$ (any J) or $I = 2b$ (necessarily with $J = 2$). Thus

$$\mathbb{P}(\text{arrival blocked at Stage 1}) = \pi_{2,0} + \pi_{2,1} + \pi_{2,2} + \pi_{2b,2} = \frac{14}{75} + \frac{2}{25} + \frac{1}{25} + \frac{2}{25} = \frac{29}{75} \approx 0.3867.$$

Expected number of busy chargers (Stage 1 + Stage 2).

- Stage 1 charger is *busy serving* iff $I \in \{1, 2\}$. (In $1b, 2b$ it is blocked, not serving.) So
$$\mathbb{E}[\text{busy at Stage 1}] = \mathbb{P}(I \in \{1,2\}) = \frac{46}{75}.$$
- Stage 2 charger is busy iff $J \geq 1$:
$$\mathbb{E}[\text{busy at Stage 2}] = \mathbb{P}(J \geq 1) = \frac{46}{75}.$$

Therefore,
$$\mathbb{E}[\text{busy chargers total}] = \frac{46}{75} + \frac{46}{75} = \frac{92}{75} \approx 1.2267.$$

Expected total number waiting in queue (Stage 1 queue + Stage 2 queue).

- Stage 1 has one waiting spot, occupied iff $I \in \{2, 2b\}$. Hence
$$\mathbb{E}[Q_1] = \mathbb{P}(I \in \{2, 2b\}) = \frac{29}{75}.$$
- Stage 2 has one waiting spot, occupied iff $J = 2$. Hence
$$\mathbb{E}[Q_2] = \mathbb{P}(J = 2) = \frac{8}{25}.$$

So
$$\mathbb{E}[Q_1 + Q_2] = \frac{29}{75} + \frac{8}{25} = \frac{53}{75} \approx 0.7067.$$

(d) λ_{eff}, L, and W

Effective arrival rate.

$$\lambda_{\text{eff}} = \lambda\,(1 - \mathbb{P}(\text{arrival blocked at Stage 1})) = 6\left(1 - \frac{29}{75}\right) = 6 \cdot \frac{46}{75} = \frac{92}{25} = 3.68 \text{ vehicles/hour}.$$

Expected number in system. Let I^* be the number of EVs physically in Stage 1:

$$I^* = \begin{cases} 0, & I = 0, \\ 1, & I = 1 \text{ or } 1b, \\ 2, & I = 2 \text{ or } 2b. \end{cases}$$

Then
$$L = \mathbb{E}[I^* + J].$$

Compute:
$$L = \frac{154}{75} \approx 2.0533.$$

Little's Law.

$$W = \frac{L}{\lambda_{\text{eff}}} = \frac{154/75}{92/25} = \frac{77}{138} \text{ hours.}$$

Convert to minutes:

$$W = \frac{77}{138} \cdot 60 \approx 33.48 \text{ minutes.}$$

Q3. Car Manufacturing

Let $X(t)$ be the total number of jobs in assembly. Jobs arrive through:

- standard mechanism at rate λ for all states,
- expedited mechanism at additional rate θ only when $X(t) \geq N$.

Each existing job departs independently at rate μ, so when $X(t) = j$, total departure rate is $j\mu$.

(a) Birth-death setup and rate diagram

States: $j \in \{0, 1, 2, \ldots\}$.

Birth rates:

$$\lambda_j = \begin{cases} \lambda, & j = 0, 1, \ldots, N-1, \\ \lambda + \theta, & j \geq N. \end{cases}$$

Death rates:

$$\mu_j = j\mu, \qquad j \geq 1, \quad \mu_0 = 0.$$

Rate diagram (conceptual):

$$0 \underset{\mu}{\overset{\lambda}{\rightleftharpoons}} 1 \underset{2\mu}{\overset{\lambda}{\rightleftharpoons}} 2 \underset{3\mu}{\overset{\lambda}{\rightleftharpoons}} \cdots \underset{N\mu}{\overset{\lambda}{\rightleftharpoons}} N \underset{(N+1)\mu}{\overset{\lambda+\theta}{\rightleftharpoons}} N+1 \underset{(N+2)\mu}{\overset{\lambda+\theta}{\rightleftharpoons}} N+2 \cdots$$

(b) Balance equations for steady-state probabilities P_j (do not solve)

Let $P_j = \mathbb{P}(X = j)$ in steady state.

Boundary equation at $j = 0$:

$$P_0\lambda_0 = P_1\mu_1 \quad \Longleftrightarrow \quad P_0\lambda = P_1(\mu).$$

For $1 \leq j \leq N-1$ (where $\lambda_j = \lambda$):

$$P_j(\lambda + j\mu) = P_{j-1}\lambda + P_{j+1}(j+1)\mu.$$

For $j = N$ (the first state whose *incoming* birth rate from $N-1$ is λ but whose *outgoing* birth rate is $\lambda + \theta$):

$$P_N\big((\lambda + \theta) + N\mu\big) = P_{N-1}\lambda + P_{N+1}(N+1)\mu.$$

For $j \geq N+1$ (where $\lambda_j = \lambda + \theta$):

$$P_j\big((\lambda + \theta) + j\mu\big) = P_{j-1}(\lambda + \theta) + P_{j+1}(j+1)\mu.$$

Normalization:

$$\sum_{j=0}^{\infty} P_j = 1.$$

(c) For $N = 10$, proportion of time only standard mechanism is active

The expedited mechanism is inactive exactly when $X < N$. With $N = 10$,

$$\text{Proportion of time expedited is inactive} = \mathbb{P}(X \leq 9) = \sum_{j=0}^{9} P_j.$$

Q4. Performance Comparison of $2 \times M/M/1$ vs $1 \times M/M/2$

Arrivals are Poisson rate λ, service times exponential rate μ.

System A: Two independent $M/M/1$ queues, random splitting

Each arriving customer chooses a queue with probability 1/2, so each queue gets rate $\lambda/2$. Each queue is $M/M/1$ with arrival $\lambda/2$, service μ. Stability requires $\lambda/2 < \mu$, i.e. $\lambda < 2\mu$.

Define $\rho_1 = \frac{\lambda/2}{\mu} = \frac{\lambda}{2\mu}$.

For one $M/M/1$:

$$L^{(1)} = \frac{\rho_1}{1-\rho_1}, \qquad W^{(1)} = \frac{1}{\mu - \lambda/2}, \qquad W_q^{(1)} = W^{(1)} - \frac{1}{\mu} = \frac{\rho_1}{\mu - \lambda/2}.$$

Total across both queues:

$$L_A = 2L^{(1)} = 2 \cdot \frac{\rho_1}{1-\rho_1}, \qquad W_A = \frac{L_A}{\lambda}, \qquad W_{q,A} = W_A - \frac{1}{\mu}.$$

(Equivalently, since the two queues are identical, the mean time in system experienced by an arbitrary arrival is $W_A = W^{(1)}$, and the same for $W_{q,A} = W_q^{(1)}$.)

Thus we can write directly:

$$W_A = \frac{1}{\mu - \lambda/2}, \qquad W_{q,A} = \frac{1}{\mu - \lambda/2} - \frac{1}{\mu}.$$

And

$$L_A = \lambda W_A = \frac{\lambda}{\mu - \lambda/2}.$$

System B: One $M/M/2$ queue with two servers

Here $c = 2$. Define

$$\rho = \frac{\lambda}{2\mu}, \qquad \text{stability: } \rho < 1 \ (\lambda < 2\mu), \qquad a = \frac{\lambda}{\mu}.$$

The empty-system probability is

$$P_{0,B} = \left[\sum_{n=0}^{1} \frac{a^n}{n!} + \frac{a^2}{2!(1-\rho)} \right]^{-1} = \left[1 + a + \frac{a^2}{2(1-\rho)} \right]^{-1}.$$

The Erlang-C waiting probability (probability an arrival must wait) is

$$P(\text{wait}) = \frac{\frac{a^2}{2!(1-\rho)}}{1 + a + \frac{a^2}{2!(1-\rho)}}.$$

Queue length in steady state:

$$L_{q,B} = P_{0,B} \cdot \frac{a^2 \rho}{2!(1-\rho)^2} = P_{0,B} \cdot \frac{a^2 \rho}{2(1-\rho)^2}.$$

Then

$$W_{q,B} = \frac{L_{q,B}}{\lambda}, \qquad W_B = W_{q,B} + \frac{1}{\mu}, \qquad L_B = \lambda W_B.$$

(a) Compute L, W, W_q for each system

System A.

$$W_A = \frac{1}{\mu - \lambda/2}, \qquad W_{q,A} = \frac{1}{\mu - \lambda/2} - \frac{1}{\mu}, \qquad L_A = \lambda W_A = \frac{\lambda}{\mu - \lambda/2}.$$

System B. Using $a = \lambda/\mu$ and $\rho = \lambda/(2\mu)$:

$$P_{0,B} = \left[1 + \frac{\lambda}{\mu} + \frac{(\lambda/\mu)^2}{2(1 - \lambda/(2\mu))}\right]^{-1},$$

$$L_{q,B} = P_{0,B} \cdot \frac{(\lambda/\mu)^2 \cdot (\lambda/(2\mu))}{2\left(1 - \lambda/(2\mu)\right)^2}, \qquad W_{q,B} = \frac{L_{q,B}}{\lambda},$$

$$W_B = W_{q,B} + \frac{1}{\mu}, \qquad L_B = \lambda W_B.$$

(b) Probability the system is empty in steady state

System A. Each queue empty probability is $1 - \rho_1 = 1 - \lambda/(2\mu)$. Independence implies

$$P_{0,A} = \left(1 - \frac{\lambda}{2\mu}\right)^2.$$

System B.

$$P_{0,B} = \left[1 + \frac{\lambda}{\mu} + \frac{(\lambda/\mu)^2}{2\left(1 - \lambda/(2\mu)\right)}\right]^{-1}.$$

To determine which is larger, compare the two expressions for any given (λ, μ) satisfying $\lambda < 2\mu$.

(c) Compare the two systems using W_q

$$W_{q,A} = \frac{1}{\mu - \lambda/2} - \frac{1}{\mu}, \qquad W_{q,B} = \frac{L_{q,B}}{\lambda}, \quad L_{q,B} = P_{0,B} \cdot \frac{(\lambda/\mu)^2 \cdot (\lambda/(2\mu))}{2(1 - \lambda/(2\mu))^2}.$$

Qualitatively (and in standard queueing theory), pooling servers into a single queue (System B) typically reduces waiting variability and yields smaller W_q than random splitting into separate queues (System A), especially as utilization increases. For a specific numeric conclusion, plug your λ, μ into the formulas above and compare $W_{q,A}$ vs $W_{q,B}$.